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“It really just wanted to talk about whales”

Addressing Misconceptions of LLMs amongst Lay Audiences

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Abstract

Lorem ipsum dolor sit amet, his veri singulis necessitatibus ad. Nec insolens periculis ex. Te pro purto eros error, nec alia graeci placerat cu. Hinc volutpat similique no qui, ad labitur mentitum democritum sea. Sale inimicus te eum.

No eros nemore impedit his, per at salutandi eloquentiam, ea semper euismod meliore sea. Mutat scaevola cotidieque cu mel. Eum an convenire tractatos, ei duo nulla molestie, quis hendrerit et vix. In aliquam intellegam philosophia sea. At quo bonorum adipisci. Eros labitur deleniti ius in, sonet congue ius at, pro suas meis habeo no.

Acknowledgements

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`Alexander`
Sunday 24th May, 2026

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2.1.1 A Brief History of AI

'Can machines think?' This question was famously posed by Alan Turing in his 1950 paper "Computing Machinery and Intelligence" [53]. Rather than trying to define what it means for a machine to think, Turing proposed the "Imitation Game", now commonly known as the Turing Test, as a way to determine whether a machine can exhibit intelligent behavior indistinguishable from that of a human. The game consists of 3 participants: a human interrogator, a human respondent, and a machine respondent. The interrogator is tasked with determining which of the two respondents is the machine, based solely on their responses to questions. The machine, on the other hand, is tasked with convincing the interrogator that it is the human respondent.

As a consequence of this, Turing shifted the discussion from the abstract question of whether machines can think based on their internal cognition, to the more practical question of whether machines can exhibit intelligent behavior through observable linguistic behaviour. This emphasis on conversational imitation would many years later be central to both the public perception and the scientific development of Artificial Intelligence (AI).

The term "Artificial Intelligence" has its origins from 1955, when John McCarthy quite optimistaclly proposed a 2 month workshop the following summer, where 10 researchers would gather to solve this "thinking machines" thing [51]. The term was created to distinguish the field from the also up and comming field of cybernetics. This early optimism is a common theme in the history of AI, with the field experiencing several periods of hype and subsequent disappointment, often referred to as “AI winters” [51]. Importantly, it has always been clear that AI has a tendency to be met with great expectations, which often also results in misunderstandings of what AI is and what it can do.

2.1.2 Exposure to the General Masses

One of the most significant milestones in the public perception of AI was the development of ELIZA by Joseph Weizenbaum in 1964 [51]. ELIZA was a simple program that mimicked a Rogerian psychotherapist, using pattern matching and substitution to create the illusion of understanding. Despite its simple inner workings, ELIZA was able to convince many users that it had genuine understanding and empathy. The phenomenon of users attributing human-like qualities to a machine, is still known as the “ELIZA effect” [51].

While ELIZA and other earlier AI systems were primarily used in research and specialized industrial settings, the public release of ChatGPT at the end of 2022 marked a significant shift in the accessibility and visibility of AI technology. This is clear by looking at google search trends for the term “AI”, shown in Figure 2.1, which shows a significant spike in interest following the release of ChatGPT. This widespread adoption of Large Language Models (LLMs) introduced new opportunities for interaction with AI systems, but also new challenges in terms of understanding and managing the expectations and perceptions of users.

This raises the question if we have failed in educating lay audiences about the capabilities and limitations of LLMs, and how to best aaddress the misconceptions that may arise from the widespread use of these systems.

2.2 Didactic Models

2.2.1 The Zone of Proximal Flow

2.3 Content Analysis

Content analysis is a research method used to analyze and interpret the content of various forms of communication, such as text, audio, or video [14]. It is a technique designed

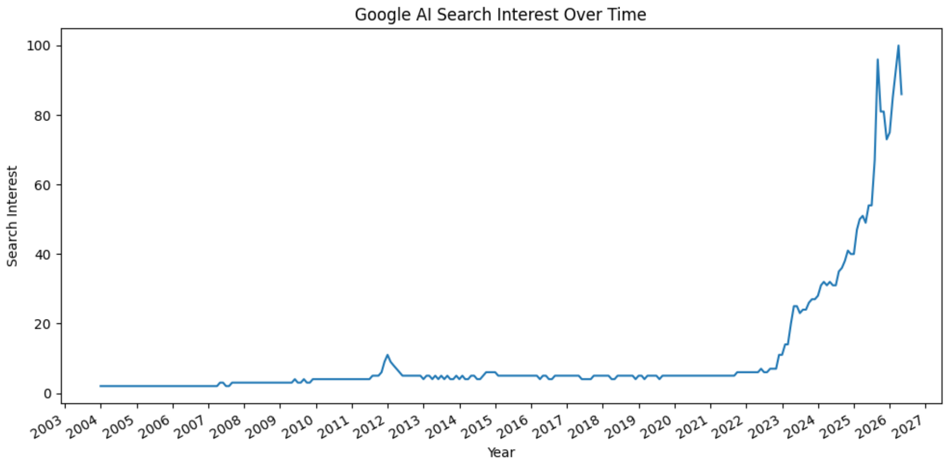


Figure 2.1: Google search trends for the term "AI" from 2004 to 2026. Data retrieved from Google Trends.

for making replicable and valid results [25]. It involves systematically describing and quantifying the content of communication in order to identify patterns, themes, and relationships. This involves coding the content into categories, which can be done using either an inductive or deductive approach. In an inductive approach, categories are derived from the data itself, while in a deductive approach, categories are predefined based on existing theories or frameworks [14]. If there are more than one coder, it is possible to calculate inter-coder reliability, which is a measure of the agreement between coders. This can be done using various coefficients, such as Cohen's κ or Krippendorff's α.

2.3.1 Deductive Approach

In a deductive approach to content analysis, the researcher starts with a predefined set of categories based on existing theories or literature [14]. After preparation, the researcher develops a coding scheme, which is a set of rules for how to assign content to the predefined categories. The researcher then applies this coding scheme to the data, systematically categorizing the content according to the predefined categories. The common coding scheme is important for ensuring consistency and reliability across multiple coders.

2.3.2 Inter-Rater Reliability

Data is used as trusted ground for making inferences, but in order to make content analysis indisputable, it is important to ensure that the data is generated with the necessary precautions and that there is a common understanding of what the data represents [25]. When multiple coders are involved in the coding process, it is important to ensure that they have a common understanding of the coding scheme and that they apply it consistently. This can be assessed through Inter-Rater Reliability (IRR) analysis, which measures the level of agreement between coders. For categorical data, Cohen's κ and Krippendorff's α are common coefficients for measuring IRR.

Cohen's κ

Cohens κ is a coefficient that measures aggreement between two raters who each classify items into mutually exclusive categories. It was first presented by Jacob Cohen in a paper published in 1960 [9]. Given a collection of categories, and 2 coders who classify items into these categories, Cohen's κ is calculated as follows:

$$\kappa = \frac{p_o - p_e}{1 - p_c} \tag{2.1}$$

where p_o is the relative observed agreement among raters, and p_c is the hypothetical probability of agreement by chance. It can be translated to the use of frequencies as follows:

$$\kappa = \frac{f_o - f_c}{N - f_c} \tag{2.2}$$

where f_o is the observed frequency of agreement, f_c is the expected frequency of agreement by chance, and N is the total number of items.

The value of κ can range from -1 to 1, where it gets a value of 0 when the agreement is exactly what would be expected by chance. A positive value indicates agreement, with a value of 1 indicating perfect agreement. Values above 0.6 indicate substantial agreement, while values above 0.8 indicate good agreement [37]. The value of 1 is only achieved when both coders classify the same number of items into each category. Therefore, you can still argue that there is agreement even if κ is quite low, by comparing the calculated κ to the maximum possible κ_M, which is calculated as follows:

$$\kappa_M = \frac{p_{oM} - p_c}{1 - p_c} \tag{2.3}$$

Where p_{oM} is the maximum observed agreement, found by summing the minimum of the two coders' probabilities for each category.

Table 2.1: Tabular representation of binary coding

	Item 1	Item 2	...	Item k
Coder 1	c_{11}	c_{12}	...	c_{1k}
Coder 2	c_{21}	c_{22}	...	c_{2k}

Cohen's κ is criticized for penalizing coders who have agreement on marginal frequencies, as two coders with marginal frequencies must have much higher agreement-rate than two coders with "radically different marginals" [25].

Krippendorff's α

Krippendorff's α is a coefficient that measures the reliability of coding in content analysis [25]. In its most basic form, it is calculated as follows:

$$\alpha = 1 - \frac{D_o}{D_e} \tag{2.4}$$

where D_o is the observed disagreement, and D_e is the expected disagreement by chance. How these are calculated depends on the data. It normally lands on a value between 0 and 1, where a value of 1 indicates perfect agreement, and a value of 0 indicates agreement and disagreement ocur by chance. It can however be negative if the observed disagreement is greater than the expected disagreement by chance, which indicates systematic disagreement [25].

Given 2 coders, a set of categories, and a set of items, you can calculate α for each category by using the binary data approach [25] on the presence or absence of each category in the coding. This process starts by tabulating the encodings for each category with one row for each coder, and one column for each item (shown in Table 2.1). This is called a reliability data matrix.

From these encodings, you can create coincidence matrices for each category as shown in Table 2.2. Here, o_{00} is the number of items where both coders did not code the category, o_{11} is the number of items where both coders coded the category, and o_{01} and o_{10} are the number of items where only one of the coders coded the category.

For binary nominal data with two coders, disagreement occurs when one coder assigns 0 and the other assigns 1. The observed disagreement is therefore proportional to the number of off-diagonal entries in the coincidence matrix.

Table 2.2: Tabular representation of coincidence matrix for a category

	0	1	
0	o_{00}	o_{01}	n_0
1	o_{10}	o_{11}	n_1
	n_0	n_1	n

Using the binary coincidence formulation presented by Krippendorff [25], the coefficient can be simplified to:

$$\alpha = 1 - (n-1)\frac{o_{01}}{n_0 n_1} \tag{2.5}$$

Chapter 3

Misconceptions Taxonomy

In order to systematically analyze the misconceptions surrounding LLMs, and the effects of the developed mediation tool, we have developed a taxonomy of misconceptions. This taxonomy is based on a review of the literature on Large Language Model (LLM) misconceptions, as well as an analysis of my own experiences and observations of LLM misconceptions. The field of LLM misconceptions is vast and rapidly evolving, but still its early stages, and there is no widely accepted taxonomy of misconceptions.

Taxonomies are classification systems aimed at helping to organize and understand complex phenomena based on characteristics [26]. Their use is widely spread in numerous fields to provide a framework for classification, idenfication and description of phenomena [48].

Since the field of LLM misconceptions is still in its early stages, we have developed a taxonomy that is based on our own analysis and observations, rather than relying on existing taxonomies. Due to the vastness of the field, we have divided the taxonomy into four dimensions, each of which captures a different aspect of LLM misconceptions. These dimensions are: foundational conceptual accuracy, mechanistic understanding, ethical and academic integrity, and socio-cognitive impact. Each dimension is further divided into subcategories, which capture specific types of misconceptions within that dimension.

3.1 Foundational Conceptual Accuracy

This dimension of the taxonomy is focused on misconceptions related to the foundational conceptual understanding of LLMs, and also the confusion between LLMs and other types

Table 3.1: Dimension 1: Foundational Conceptual Accuracy

Misconception	**Label**	**Description**
Anthropomorphism	ANTHRO	Attributing human mental states, intentions, or understanding to LLMs.
Deterministic Automation	DETAUTO	Confusing LLMs with deterministic automation systems.
Accuracy Overestimation	OVERACC	Overestimating the reliability and correctness of LLM outputs.
Autonomy Overestimation	OVERAUTO	Belief that the LLM operates independently or possesses agency.

of AI systems. The term "AI" exploded in popularity after the relase of GPT-3.5 model, and the term is often used to refer to a wide range of technologies, including LLMs. This has lead to a lot of confusion and misconceptions about what LLMs are and what they can do.

3.1.1 Anthropomorphism

The category of anthropomorphism is defined as the attribution of human mental states, intentions, or understanding to LLMs. This misconception seems to be common amongst users who interact with LLMs in a conversational manner, like chatbots. Subjects was placed in this category by using terms such as "understand", "think", "learn" and "know" to describe the LLM's capabilities in a non-metaphorical way.

Anthropomorphism emerges both from the system design of LLMs, and from the user's interpretation of the system's behavior [32]. A paper by Maeda and Quan-Haase provides a framework for when Human-AI Interaction (HAI) become parasocial. They argue that chatbot's variety in information presentation resemble day-to-day bi-directional communication, and that the users' tendency to fill in missing conversational context causes them to attribute human-like qualities to the system. The roleplaying nature of the chatbot interface is visible through language choices such as personal pronouns and paraphrasing suggests attributes such as intention and understanding in the system which lead users to project inference to the system due to the lack of information about the system's inner workings [32]. This is empirically supported by a large scale study by Ibrahim et al. which found similar anthropomorphic behaviours in all tested LLMs (Gemini 1.5 Pro, Claude 3.5 Sonnet, GPT-4o and Mistral Large) [22]. They found, in particular, use of personal pronouns, validation and empathy as the most common anthropomorphic behaviours, and that these behaviours mostly occurred after multiple turns of conversation.

The anthropomorphism misconception is widespread. An investigation by Colombatto and Fleming in 2024 found that in a sample of 300 US residents, only a third of the participants thought that ChatGPT definitely did not have subjective experiences, while two thirds applied varying degrees of phenomenal consciousness [10]. The investigation also showed that the participants' conciusness attributions increased with usage frequency, rather than correcting the misconception.

Building on this, Lin [29] conceptualizes anthropomorphism as an "anthropomorphism fallacy", in which users and researchers misinterpret LLM outputs as evidence of underlying mental life. The fluent, human-seeming responses of LLMs can lead to an enhanced ELIZA effect, encouraging users to attribute understanding, beliefs, or intentions to systems that in reality operate through statistical token prediction. This creates a "language-user illusion", where the system's linguistic capabilities are mistaken for genuine cognition.

Furthermore, the anthropomorphic metaphors used surrounding LLMs can reinforce this misconception [29]. Unlike humans who are influenced by their own experiences, LLMs simulate a wide range of viewpoints based on their training data [29], which can lead to overestimation of their capabilities.

3.1.2 Deterministic Automation

The category of deterministic automation is defined as the confusion between LLMs and deterministic automation systems. One could say that this misconception is placed on the oposite side of the spectrum compared to the previous misconception of anthropomorphism. Before the widespread use of LLMs, one could often find deterministic automation systems in customer service chatbots, which were designed to follow a fixed set of rules and provide predetermined responses. It seems natural that users would initially confuse LLMs with these deterministic systems, especially since they are often presented in similar interfaces. Subjects were placed in this category if they described LLMs as following a fixed set of rules, or if they expressed surprise at the variability and creativity of LLM outputs.

This misconception is likely reasoned by a lack of interaction experience with LLMs. In a study performed by Schneider in 2025, analyzing over 200 000 conversations, he found that users initially approached LLMs as traditional software tools [47]. Initial prompts were often structured and machine-like, but already on the second turn, users shifted towards more natural, conversational interaction. Thsi was shown by the use of a less complex language, use of politeness indicators, shorter prompts and more personal interaction.

A concequence of this misconception comes when LLMs actually replace deterministic automation systems, leading to failures in safety-critical deployments. Vinay (2025) argues that LLM failure patterns "differ fundamentally from those of traditional machine learning models" and that a lack of awareness of these differences can lead to failures going undetected [56]. For developers, automotive code generation tools produce syntax violations, invalid refrence errors and Application Programming Interface (API) knowledge conflicts in high rates, which are failures deterministic systems do not produce [39].

This misconception could also work as a springboard to other misconceptions covered in this taxonomy. Patterns of logical reasoning could lead to an overtrust in LLMs leading to placements in categories of OVERACC and CREDBIAS [16, 34].

3.1.3 Accuracy Overestimation

The category of accuracy overestimation is defined as the overestimation of the reliability and correctness of LLM outputs. This misconception is likely corelated with other misconceptions in this taxonomy, on which failure to understand fundamental concepts and mechanics of LLMs can lead to an overestimation of their capabilities. Subjects were placed in this category if they expressed a belief that LLMs are more accurate or reliable than they actually are, or if they failed to recognize the potential for errors and inaccuracies in LLM outputs.

The most fundamental reason for this misconception is automation bias, which is the tendency for people to trust in automated systems even when they contradict available evidence. Carnat (2024) argues that this phenomena spawns from ANTHRO [4]. The misconception is widespread, shown by a experiment by Klingbeil et al. in 2024, where they found that the mere knowledge of advice being generated by an AI causes people to overrely on it. This is the case even when the advice contradicts available contextual information and their own assessments [24].

Accuracy overestimation is even present in the LLMs themselves. In 2025, Sun et al. found that after testing the confidence calibration of five different LLMs, they overestimate their accuracy by 20% to 60% [52]. Naderi et al. also found similar results saying that LLMs retained high confidence regardless of the correctness of their outputs [33].

A rather obvious consequence of this misconception is the spreading of misinformation. The logical patterned output of LLMs are often perceived to be valid, trustworthy and satisfactory, and users also show a high tendency to follow LLM advice provided [49].

3.1.4 Autonomy Overestimation

The category of autonomy overestimation is defined as the belief that the LLM operates independently or possesses agency. This category was was inspired by a study performed by Wang et al. in 2025, where they researched the mental models of Generative Artificial Intelligence (GenAI) chatbot ecosystems [57]. In this study, participants where to asked to draw a diagram of how they thought the chatbot ecosystem worked when they used it to book a room on a hotel. The study showed various mental models of the degree of autonomy of the GenAI in the ecosystem. Participants placed in this category expressed beliefs that the LLM operates independently, or "acts" on behalf of the user.

It is likely that this misconception spawns from the ANTHRO misconception, as it is natural to assume a real social actor when presented with a system that produces natural language outputs[42, 23]. Thus, it is expected to be a high level of corelation between these two categories.

3.2 Mechanistic Understanding

The second dimension of the taxonomy is focused on misconceptions related to mechanics specific to that of a LLM. Most modern LLMs today are transformer-based models, and not being aware of the mechanics of how these models work can lead to misconceptions about their capabilities and limitations. This dimension differs from the first dimension of foundational conceptual accuracy, as it is focused on specific mechanics of LLMs, rather than that of AI in general.

3.2.1 Real-Time Data Belief

LLMs are trained on large, fixed datasets, and generate responses based on patterns learned from that data. Therefore, mistakenly believing that LLMs can access real-time information or the internet is a common misconception. People placed in this category

Table 3.2: Dimension 2: Mechanistic Understanding

Misconception	**Label**	**Description**
Real-Time Data Belief	REALTIME	Belief that LLMs retrieves current information
Hallucination Blindness	HALLBLIND	Failure to recognize that LLMs can produce false or fabricated information.
Context Window Neglect	CONTEXNEG	Failure to understand the limitations of the context window
Semantic Understanding	SEMUNDER	Lack of awareness that the LLM processes language as tokens rather than semantic units.
Source Attribution Error	SOURCEATTR	Belief that LLM output originate from specific identifiable sources.
Conditional Treatment	CONDTREAT	Belief that LLM process prompts differently based on the content or context

expressed beliefs that LLMs will gather information from the ongoing conversation and that will "update" the model's knowledge.

Live access to the internet is also under the category of real-time data belief, as LLMs by themselves do not have the capability to access the internet or retrieve real-time information. The introduction of Retrieval-Augmented Generation (RAG) [27] has somewhat blurred the lines here, as it allows LLMs to retrieve information from external sources, but it is still a misconception to believe that LLMs have this capability by default. It is still necesary to be aware of the limitations of RAG systems, as it is still limited to the underlying LLM's "understanding" of the retrieved information [58].

3.2.2 Hallucination Blindness

The term "hallucination" in the context of LLMs refers to when the output generated by the model is deemed as factually incorrect. Many believe that this is an error done by the LLM, and that is what we hace categorized as hallucination blindness, which is the failure to recognize that LLMs can produce false or fabricated information.

A LLM is trained by recognizing syntactic patterns in the training data, and it generates responses based on those patterns. The semantic value of natural language is in the end evaluated by the receiver of the information, and there is no way in the LLM's design to interpret such a semantic value. Therefore, a hallucination done by a LLM is actually not an error, but rather a feature of how the model operates.

Failure in realizing that hallucination is, and will continiue to be, a feature of LLMs may cause users to be categorized into other misconceptions such as OVERACC or CREDBIAS. There is an argument to be made that the use of RAG will mitigate generation of incorrect information, but in order to have a model able to generalize, it is still necessary to have a model that can generate information based on patterns, which will still lead to hallucination.

3.2.3 Context Window Neglect

LLMs operate within a fixed sized context window, which limits the amount of information they can process at once. That is, you can only input a certain amount of tokens into the model. Subject who showed no indication of understanding this limitation, or who expressed beliefs that LLMs can process unlimited amounts of information, were categorized into context window neglect.

Even within the context window, there is limitations to how much information the model can effectively utilize. According to a study performed by Liu et al. in 2023, LLMs shows much higher performance if the relevant information is placed at the beginning or the end of the input context [30]. Even if all the relevant information is retrieved correctly from the context, LLMs performance still degrades noticably as the amount of information increases [13]. A study done by Paulsen in 2025 showed that given input of datasets of varying sizes, the top-line LLMs failed to provide corect answers when the context dataset size exceeded 100 tokens, and that the accuracy plummeted when the context dataset size exceeded 1000 tokens [38].

Failure to understand the limitations of the context window could be strongly corelated with other misconceptions in this taxonomy, such as OVERACC, REALTIME and CREDBIAS, as it is likely that users who do not understand the limitations of the context window will also overestimate the accuracy of LLM outputs, and believe that LLMs can access real-time information. This corelation seems to common amongst students using LLMs for academic purposes [36, 44].

3.2.4 Semantic Understanding

LLMs process language as tokens rather than semantic units. This means that the model does not understand language in the way humans do, but rather as a sequence of many-dimensional vectors, which values are used to calculate the probability of the next token in the sequence. This is done through first splitting the input text into tokens, which can be done on different levels, such as words, subwords or even characters. Most modern LLMs use subword tokenization to capture smaller units of meaning while still being able to handle words not in vocabulary.

After the tokenization process, the tokens are converted into embeddings through a pre-trained word embedding model. A word embedding model is a machine learning model that is trained to represent words as vectors in a high-dimensional space, where semantically similar words are represented by similar vectors. The idea is that the model can capture the semantic relationships between words based on their co-occurrence in the training data. For example, the distance between the vectors for “king” and “queen” would be similar to the distance between the vectors for “man” and “woman”, reflecting the semantic relationship between these words.

Understanding text semantically is a human ability which cannot be fully captured by a statistical model. Therefore, it is a misconception to believe that LLMs understand language in the same way humans do. Subjects who expressed beliefs that LLMs have a semantic understanding of language, or who failed to recognize that LLMs process language as tokens, were categorized into semantic understanding. There should be a high level of corelation between this misconception and the ANTHRO misconception, as it is likely that users who anthropomorphize LLMs will also believe that they have a semantic understanding of language. Since the output of LLMs also are embeddings turned back into tokens, it is likely that users who do not understand the tokenization process will also fail to understand that the output is also generated based on token patterns rather than semantic understanding, which could lead to placements in the OVERACC and CREDBIAS categories.

3.2.5 Source Attribution Error

Since LLMs are trained to mimic the patterns in their human-written training data, it comes as no surprise that the generated output will resemble how humans write and cite their sources. The generated output of LLMs is still limited to the statistical patterns in the training data, and even if the output resembles a citation, it does not mean that the LLM is actually retrieving information from specific identifiable sources. For RAG systems, the retrieved information is still processed based on token patterns, and the LLM does not have an understanding of the information it retrieves. One could argue that the use of RAG systems does make the citation identifiable and thus the information is retrieved from specific identifiable sources, but the information is still processed based on token patterns.

Once the LLMs have gone through training, it is also impossible to trace back where the generated information comes from in the training data, as the model does not store information in a way that allows for such tracing. Subjects who in some way expressed belief that LLM output originate from specific identifiable sources, or to some degree expressed that the information is learned in training, were categorized into source attribution error.

The mimicing of citation patterns by LLMs can lead to an overestimation of the reliability of the generated information and therefore lead to placements in the OVERACC category. A study by Ding et al. in 2025 found that users deemed responses with citations as more reliable than those without, even if the citations were randomly generated. Fact checking these citations lead to a significant decrease in trust in the response [11]. This result is backed by Do et al. who found that output with citations were deemed more trustworthy than output without [12].

3.2.6 Conditional Treatment

Every prompt given to a LLM is processed based on the same underlying mechanics, which is the statistical patterns in the training data. The length, or the semantic complexity of the prompt does not change the underlying mechanics and therefore it is a misconception to believe that LLMs process prompts differently based on the content or context. Subjects who in some way expressed belief that LLMs will process one prompt differently than another were placed in this category.

3.3 Ethical and Academic Integrity

The third dimension of the taxonomy shifts focus away from the what and how of LLMs, and instead focuses on issues related to the ethical and academic integrity of LLM use. The use of LLMs in academic settings has raised a lot of ethical questions, and there is an ongoing debate on whether LLMs has a place in academia at all, and if so, under what conditions. There are also ethical questions related to the use of LLMs outside of academia, such as the environmental impact of training and operating LLMs.

Table 3.3: Dimension 3: Ethical and Academic Integrity

Misconception	Label	Description
Plagiarism Blindness	PLAGBLIND	Failure to recognize ethical and legal issues related to the traceability of sources behind LLM-generated content.
Copyright Blindness	COPYBLIND	Failure to recognize that LLM-generated content may still be subject to copyright and usage restrictions, despite being machine-generated.
Conditional Acceptance	CONDACC	Ethical acceptance of LLM use only under certain conditions or domains.
Environmental Neglect	ENVNEGL	Failure to recognize the environmental impact of LLM training and operation.

3.3.1 Plagiarism Blindness

A core idea in academia is that ideas and information should be traceable to their sources, and that proper attribution should be given to the original creators of those ideas. This is imortant for maintaining academic integrity and giving credit where it is due. However, since there is no way to trace exactly where the generated information comes from in the training data, it is a misconception to believe that LLM-generated content is free from ethical and legal issues related to plagiarism. Subjects who showed little to no awareness of the ethical and legal issues related to the traceability of sources behind LLM-generated content were categorized into plagiarism blindness.

A common usecase of LLMs is to generate ideas for starting a project. At the Faculty of Social Sciences at The University of Bergen (UiB) they have adopted guidelines for the use of LLMs in academic work [54]. These guidelines are divided into five categories, where category 1 is the most strict with no use of LLMs allowed, and category 5 is the most lenient with no restrictions on the use of LLMs. The second most strict category, category 2, states that LLMs can be used for idea generation and information search. This seems troublesome in regards to plagiarism, as the generated ideas and information cannot be traced back to their sources in the training data. The fact that the faculty has adopted these guidelines, could contribute the the lack of awareness surroundning these ethical questions. Naithani suggest that both responses and ideas generated by a LLM should be attributed to the LLM as the author, but also raises the question of whether LLMs has the moral right to attribution [35].

3.3.2 Copyright Blindness

This misconception differs from plagiarism blindness, as it is focused on the ownership of the generated content, rather than the traceability of sources. Multiple juristrictions have already concluded that GenAI output does not meet the requirements for copyright protection, as copyrightability requires human creative involvement [15, 31]. This raises the question of who actually owns the generated content of the LLM itself, the user who provided the prompt, the developer of the LLM or the author of the training data. Subjects who failed to debate the ownership of LLM-generated content, or who expressed beliefs that LLM-generated content is not subject to copyright and usage restrictions, were categorized into copyright blindness.

3.3.3 Conditional Acceptance

Somewhat connected to the category of conditional treatment, where users express beliefs that LLMs process prompts differently based on the content or context, this can develop to the category of conditional acceptance, where this belief causes difference in use. For example, some users may express that they are comfortable using LLMs for certain tasks, such as idea generation or information search, but not for others, such as writing a paper or generating code. In the academic setting, this can also be expressed as acceptance of LLM use only in certain domains. This could be for instance claiming that LLMs are better at programming rather than physics, or any other field. This is possibly backed by data from a Statista global survey, stating that only 16% expressed belief that AI adoption would impact their job for the worse, while twice as many (36%) expressed belief that AI adoption would negatively impact the job market [7]. Subjects who expressed beliefs that they are comfortable using LLMs for certain tasks but not for others, or who expressed acceptance of LLM use only in certain domains, were categorized into conditional acceptance.

3.3.4 Environmental Neglect

3.4 Socio-Cognitive Impact

The fourth and final dimension of the taxonomy is focused on misconceptions related to the socio-cognitive impact of LLMs. The widespread use of LLMs has raised concerns about their potential impact on human cognition, and society as a whole. Some of these concerns are warranted, while others are based on misconceptions about how LLMs work and their capabilities.

Table 3.4: Dimension 4: Socio-Cognitive Impact

Misconception	**Label**	**Description**
Replacement Fear	REPLFEAR	Belief that LLMs will replace human roles, expertise, or professions.
Cognitive Passivity	COGPASS	Tendency to rely on LLMs rather than engaging in independent reasoning or critical thinking.
Dependency Risk	DEPRISK	Failure to recognize that extensive can lead to long term reliance on LLMs, skill degradation or loss of autonomy.
Credibility Bias	CREDBIAS	Treating LLM outputs as inherently truthful or authoritative without critical evaluation.

3.4.1 Replacement Fear

It remains to be seen how LLMs will impact the job market, but there is belief amongst some that LLMs will replace human roles, expertise, or professions. Multiple surveys have shown that about a third of the population express fear of job replacement by AI [7, 28]. However, more then half of US workers found it unlikely that their job would be replaced by AI, but rather change how they do their current job [28].

Where this fear becomes unwarranted, and where a misconception arises, is when people shifts from “LLMs can replace certain tasks” to “LLMs can replace human roles”. Subjects who expressed this belief was placed in the category of replacement fear.

3.4.2 Cognitive Passivity

This misconception is defined as the tendency to rely on LLMs rather than engaging in independent reasoning or critical thinking. This misconception was not categorized based on the subjects’ beliefs about the capabilities of LLMs, but rather based on their responses on how they themselves describe their own use of LLMs.

That raises the question of what sort of habits and behaviors that would be categorized into cognitive passivity. The habits most likely to create cognitive passivity are the ones that make the LLM a first-resort for information and problem-solving, and as a subsitute for thinking about the problem at hand. That is, if a user descibes that when they encounter a problem, they start of by asking a LLM for help, or if they descibe immediate use of LLMs when comparing different approaches.

A study performed by Gerlich in 2024 found that there is a high level of negative correlation between frequent use of AI-tools and critical thinking skills [17]. The mediator of this correlation was found as the increase in cognitive offloading. When users were introduced to AI-tools might be critical, as the study showed that younger participants were more likely to offload cognitive tasks to the AI-tool, which resulted in lower scores in critical thinking. However, the study also showed that a higher level of education was a protective factor against this effect, as higher educated participants were more likely to cross-check and evaluate responses from the AI-tool.

3.4.3 Dependency Risk

While the previous misconception of cognitive passivity is assigned based on the subjects' descriptions of their own use of LLMs, the category of dependency risk is assigned based on the subjects' ability to discuss the potential long-term consequences of extensive LLM use. Subjects who, in describing their use of LLMs, showed no to little recognition of the risk of becomming reliant on LLMs, or described tasks that should not require LLM use, but still used them for those tasks, were categorized into dependency risk.

As discussed in the previous category, there is a negative corelation between frequent use of AI-tools and critical thinking skills [17], and this becomes more coherent when it gets to the point of cognitive offloading. Cognitive offloading is the act of shifting tasks with cognitive demands onto external tools. There has been shown an impairment between cognitive offloading and confidence calibration, while the most accurate confidence calibration is found amongst unaided decision makers [6]. This springs into other potential consequences of dependency risk, such as spread of misinformation as a result of OVERACC.

3.4.4 Credibility Bias

This misconception is based on older automation bias research, which shows a over-reliance on automated systems, or a higher trust in automated systems [19]. If LLMs are used as decision support systems, there is a risk that this automation bias will be transfered to LLMs, leading to a credibility bias, which is the tendency to treat LLM outputs as inherently truthful or authoritative without critical evaluation. Subjects wo in someway that they treat LLM outputs as generally more truthful or authoritative than other human sources of information, or who showed no to little critical evaluation of LLM outputs, were categorized into credibility bias.

Chapter 4

Methodology

4.1 The Mediation Tool

In order to address potential misuse of LLM-tools, we wanted to develop an interactive mediation tool aimed at teaching users of the inner workings of LLMs. The intention was that by providing users with a viable mental model of how LLMs work, they would be better equipped to use them in a responsible manner. We wanted to create an interactive tool that would engage users in a way that would be effective in showing them how the different components and aspects of LLMs work together to produce the output they do. At the same time, the tool should be accessible to a wide range of users, not relying on prevoius technical or theoretical knowledge in Computer Science (CS) or AI.

The intention was that participants in the study would use the tool to learn about LLMs in their own time and at their own pace, without the need for instruction or guidance from us. Thus, it needed to be intuitive and engaging enough to encourage users to explore and learn on their own, keeping them inside the zone of proximal flow [1]. We settled on a web-based tool containing a "build your own LLM"-type experience, where users would be able to themselves choose and configure different components of a LLM and see how their choices would affect the output of the model. The technical implementation of the tool was done as a part of the Informatics Project II course at UiB, independent of this study, with David Grellscheid as supervisor.

4.1.1 Features

The Dynamic Model

The tool consists of various ways to interact with a very simple representation of a LLM. This model is modular, letting the users themselves choose which components and features they want to add to their model configuration. The model itself is a simple mapping between a token or sequence of tokens and the probability distribution of the next token. There are different versions of this mapping, depending on which features the user has selected in their configuration. There are 3 different tokenizers, 3 different context window sizes and a free choice of up to 2 out of 9 different books from Project Gutenberg [21]. The model also contains a switch for whether the model should be deterministic by selecting the most probable token as the output, or non-deterministic by sampling from the probability distribution.

Build Your Own LLM

The core feature of the tool is the "Build Your Own LLM"-experience. In this view, users will be able to interact with a Graphical User Interface (GUI) containing a prompt-bar and a skill-tree of different components and configurations that they can choose to add to their model configuration. The prompt-bar will allow users to input prompts and then see the output of their model configuration in real-time, showing them the prediction for the next token as ghost text. The skill-tree displays all selected features in the current configuration, as well as all available features that can be added to the configuration. The features are organized in a tree structure, in order to demonstrate the dependencies between different components and configurations. For example, you cannot select the 3-context-window feature without first selecting the 2-context-window feature.

Text Generator

At any point in the "Build Your Own LLM"-experience, users can explore their current configuration in a text generation context. By clicking the "Text Generator"-button, users are taken to a different view containing a larger text input field and a button saying "Generate Text". Here, users can input longer prompts and at any point click the "Generate Text"-button to have their current model configuration continously generate

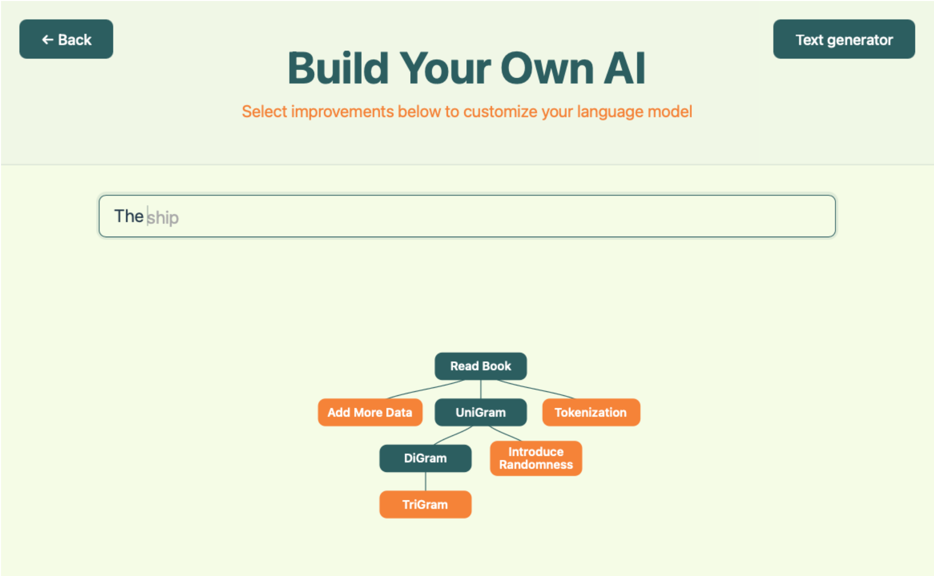


Figure 4.1: The "Build Your Own LLM"-view of the mediation tool.

the next token in the text, like a normal LLM would. This also demonstartes to the users that the same underlying process can be used for multiple purposes, and that the different features they have selected in the skill-tree work together to produce the output of the model.

Scrollytelling Chapters

Connected to each feature in the skill-tree is a scrollytelling chapter that explains the feature in more detail. Scrollytelling is a form of interactive storytelling where the content visible on the screen changes as the user scrolls down the page. This is utilized in the mediation tool to provide users with a more engaging and interactive way to learn about the different features and components of LLMs. Each chapter consists of a collection of textboxes, to each of which have a coresponding visual animation that is triggered when the textbox is scrolled into view. The animations are designed to visually demonstrate the concepts explained in the textboxes, providing users with a more intuitive understanding of how the different features and components of LLMs work together to produce the output they do.

Chapter Overview

Users can at any point access each of the individual scrollytelling chapters through a chapter overview page. This page contains a list of all the features available in the tool,

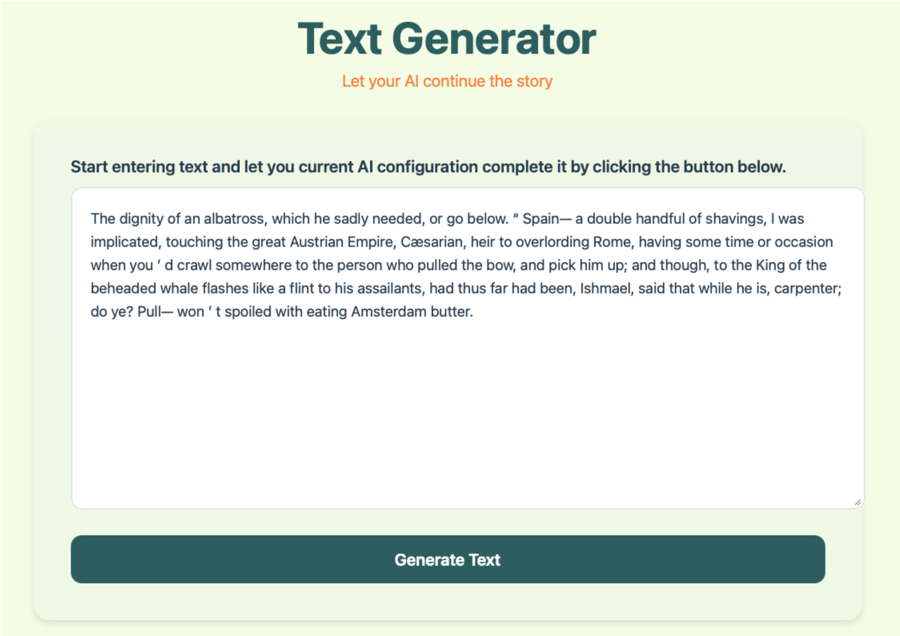


Figure 4.2: The “Text Generator”-view of the mediation tool.

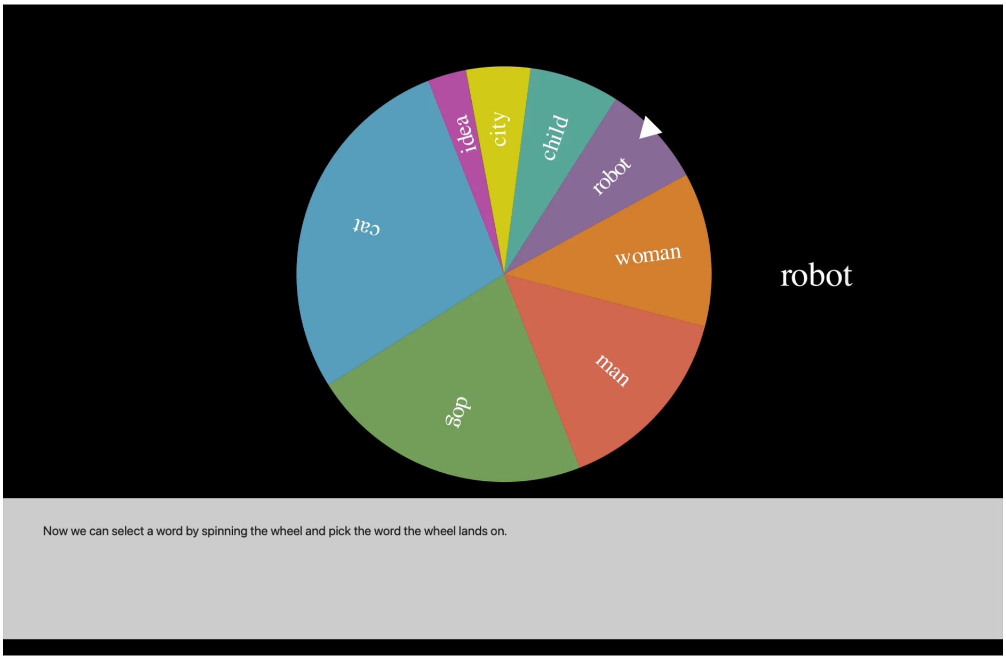


Figure 4.3: The scrollytelling chapter connected to the weighted random sampling feature.

organized by category. Each feature has a short description, and by clicking on the feature, users are taken to the corresponding scrollytelling chapter where they can learn more about the feature and how it works.

External Resources

At the bottom of the chapter overview page, there is a section containing a list of external resources for users who want to learn more about LLMs and how they work. This includes online courses, educational videos, articles and blog posts on relevant topics [2, 8, 20, 41, 45, 46, 50].

4.1.2 LLM Representation

The mediation tool does not actually contain a real transformer-based LLM under the hood, but rather a simple representation of one. Each branch of the skill tree corresponds to a different feature or component of LLMs. Most of the topics contain multiple chapters building on each other, to keep users in the zone of proximal flow [1]. The features are grouped into 4 different categories: training data, context, non-determinism and tokenization. Each of the categories are aimed at giving intuition on how the components work rather than being technically accurate, and the different features are designed to demonstrate the effect of that component on the output of the model. The selection tree, with corresponding topic, can be seen in Figure 4.4. The components selected for mediation are mainly chosen to address misconception in the dimension of Mechanistic Understanding, in hope that by giving users a better understanding of how LLMs work, they will be better equipped to use them in a responsible manner.

Training Data

This category is perhaps the most true to real LLMs, as the features in this category actually modifies the training data of the model. By understanding training data, you get an understanding on how LLMs are somehow fixed and does not gain new knowledge with interaction from the user. Thus, it will prevent the REALTIME misconception. Also, by understanding that the tokens used for training have no tracing to were in the training data they come from, it will prevent the SOURCEATTR misconception.

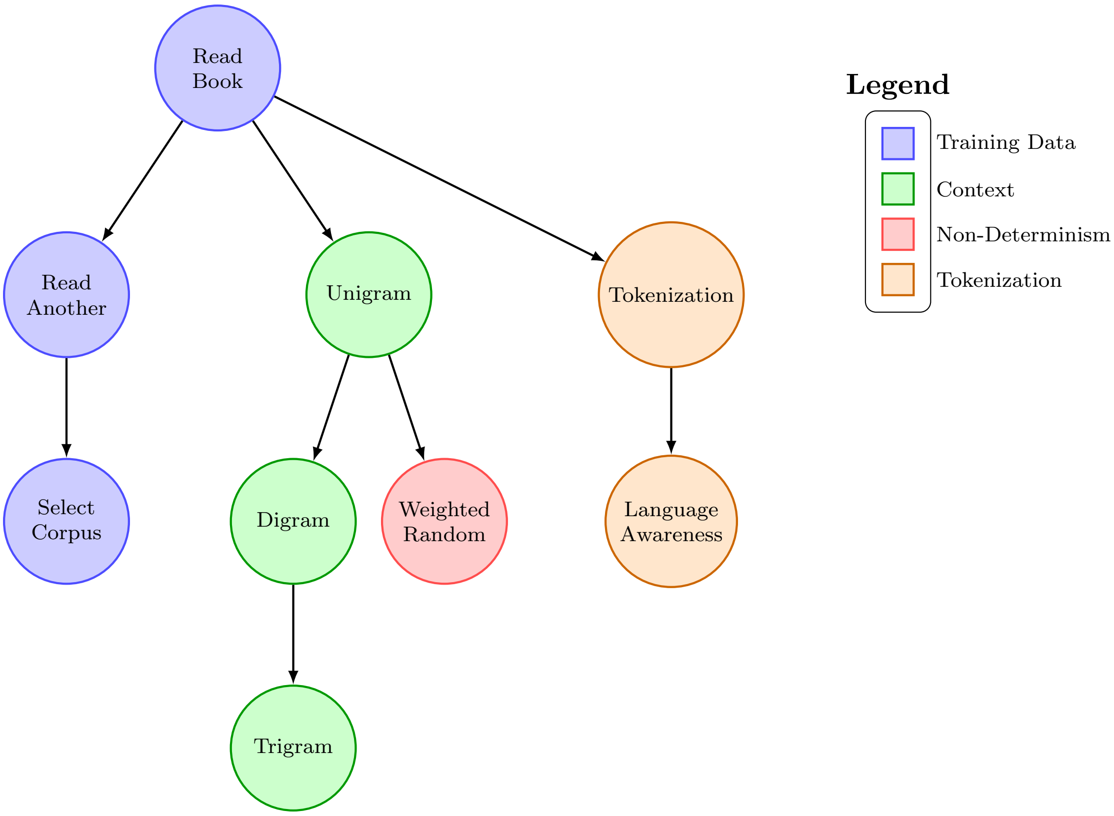


Figure 4.4: Selection tree with grouped LLM features

The first feature the users can add is the "Read Book"-feature, which simply adds a book from Project Gutenberg [21] to the training data of the model. Before adding this feature, the model works by randomly selecting a a word from the entire english language as the next token. After adding the feature, the model will only select words from the book that was added to the training data. The learning target of this feature is to introduce the idea that models are trained on previously written human text, in hope that it will learn the patterns and structures of human language.

Secondly, users can add the "Read Another"-feature, which adds a second book to the training data. This will concattinate the two books together, and the model will select the next token from either of the two books. The learning target of this feature is to show that to get LLMs to generalize, they need to be trained on a large and diverse dataset, containing many different examples of human language.

Finally, users get the option to selects which books they want to add to the training data with the "Select Corpus"-feature. This feature does not alter the underlying process of the model, but it allows users to explore how different training data can affect the output of the model. All books available in the tool are from Project Gutenberg [21].

Context Window

The context window of a LLM is the number of previous tokens that into account when predicting the next token. Understanding that this is a fixed size window and that the tokens within this window work together to produce the output of the model is important when utilizing LLMs in a responsible manner. This set of features is aimed at preventing the CONTEXNEG misconception.

In real transformer-based LLMs, the context window is implemented through the a self-attention mechanism, which is a function that takes in the previous tokens and calculates a weighted sum of them to produce the output [55]. This is a complex process, which we deemed to be too technical for the target audience of the tool. Thus, we needed a viable representation of this attention mechanism that gives understanding on how this is a statistic evaluation, and that the other tokens in the context window work together to produce the output of the model.

In order to find a viable representation of the context window, we went back to the origins of Natural Language Processing (NLP). In the mid 20th century, Claude Shannon proposed using Markov chains to model the statistical properties of language, this is what we now call n-gram models [18]. These models work by using token frequency counts to calculate the probability distribution of the nth token given the previous n-1 tokens. By introducing this model to users, we can give intuition on how the context window works, and how the tokens within this window work together to produce the output of the model.

We provided users with three different context window features: unigram, digram and trigram. Each of the features was aimed at different aspects of the context window. The unigram feature starts of by showing we can use the frequency of individual tokens to predict the next token, without taking into account any previous tokens. The digram feature shows that looking at tokens in context may change the interpretation of them, and thus give a more varied output. Finally, the trigram model illustrates that by looking at a larger context window, the model will be able to produce more coherent and contextually relevant output.

Non-Determinism

An important aspect of LLMs is that they are non-deterministic, meaning that the same prompt can produce different outputs, which is done to make more varied output. The addition of this variability comes with the cost of unpredictability and also the risk of generated output not being true to the training data. Understanding the nature of how this randomness is introduced into the model is important for responsible use of LLMs. We added this feature to prevent the HALLBLIND misconception.

In the tool, this is implemented through a simple switch which toggles between a deterministic model, which always selects the most probable token as the output, and a non-deterministic model, which samples from the probability distribution of the next token. The sampling is done through a weighted random sampling, which means that tokens with higher probabilities are more likely to be selected as the output, but there is still a chance for less probable tokens to be selected. It is visualized in the tool by showing the probability distribution as a bar chart morphing into a pie chart, which then turn into a roulette wheel that spins and lands on the selected token.

A planned feature that was not implemented in the final version of the tool was a temperature slider, which would allow users to adjust the level of randomness in the model. This was planned to be implemented in a similar manner as it is in in modern LLMs, with a softmax function (4.1) that adjusts the probability distribution of the next token based on the temperature value.

$$Softmax(z_i) = \frac{e^{z_i/T}}{\sum_j e^{z_j/T}} \tag{4.1}$$

Here z_i is the logit for token i, that is the the probability of token i being the next token before applying the softmax function. T is a temperature value that controls the level of randomness in the model. The plan was to set this value as a slider between 0.1 and 2, where a lower value ($T < 1.0$) would result in more deterministic behavior, while a higher value ($T > 1.0$) would flatten the probability distribution and make the model more random. A value of $T = 1.0$ would leave the probability distribution unchanged. This would have allowed users to explore how adjusting the level of randomness in the model can affect the output, and explain how this can be tweaked to a more convincing balance of accurate and creative output, depending on the use case.

Tokenization

The tokenization process is how the input text is converted into a format that the model can work with. This is done before it is fed into the model, and it is an important aspect of how LLMs work. LLMs needs a numerical representation of the input text in order to do the calculations needed to produce the output, and tokenization is the process of converting the input text into this numerical format. First by breaking the input text into smaller units called tokens, and then by converting these tokens into numerical representations in the form of word embeddings. We wanted the user to understand that LLMs do not understand, nor see, language in the same way as humans do and therefore should not be expected to respond to prompts in the same way as a human would. Understanding this will prevent the SEMUNDER misconception.

In the tool, the model used is not reliant on a numerical representation of the input text, but rather works directly with the tokens. Thus, we needed to find a viable representation of the tokenization process that gives users an understanding of how the input text is broken down into smaller units, and how these units are then used by the model to produce the output. The first step, is to understand that the entire input text is not processed as a whole, but broken down into tokens. The intuition given here is that the model already looks at the input text as a collection of words, but that this is not necessarily the best way to break down the input text for the model to work with. We use punctuation as an example of how the word-based tokenization can be problematic, as it will treat the same word as different tokens depending on the punctuation used with it. Thus, the first feature in this category will change from a whitespace-based tokenization to a punctuation-based Regular Expression (REGEX) tokenization.

The second step is to understand that the tokens are not just randomly selected based on the whitespace and punctuation in the input text, but that are designed to be somewhat language-aware. The syntax of natural language is complex and can be ambiguous, and this ambiguity must be captured in order for the model to be able to produce reliable output. We try to give users an intuition on this by using the Natural Language Toolkit (NLTK) library [3]. This library contains a collection of tools for working with natural language data, including a tokenizer that is designed to be language-aware.

Discussion was had on whether to include an additional feature focused on the design of word embeddings, discussing how this can include biases in the training data and how this can affect the output of the model. However, we did not find a viable way to represent this in the tool, and thus it was not included in the final version.

4.2 Study Design

4.2.1 Recruitment

Participants in the study were recruited through a Microsoft Forms survey, which was distributed through various channels. The survey was shared to students and teaching assistants in the CS1-equivalent course INF100 at UiB, as well as posted on forums of voluntary student organizations, and posters were put up around campus. To appeal to as many potential participants as possible, a lottery of cinema tickets was used as an incentive for participation. The recruitment targeted students with varying levels of technical background in order to capture a diverse range of understandings and misconceptions about LLMs.

The survey contained a brief description of the study, and participants were asked to provide their email address if they were interested in participating. The survey also contained a few questions about the participants' background including age group, occupation, and if they were a student, which institution they were affiliated with.

A total of 10 participants were recruited for the study, all students from various institutions in Bergen.

4.2.2 Assessment Interviews

The study started with an assessment interview, where participants were asked about their background and experience with LLMs, as well as their understanding of how LLMs work and their opinions on the technology. The interview was semi-structured, with a brief interview guide to ensure that relevant topics were covered, while still allowing for participants to express their thoughts and opinions in their own words. Follow-up questions were asked based on the participants' responses to explore if they contained specific misconceptions about LLMs. All interviews were conducted in Norwegian. The interview guide can be found in Norwegian in Appendix A. The interviews lasted between 15 and 45 minutes, with an average of around 30 minutes.

After the interviews were conducted, they were transcribed using the built-in transcription tool in Microsoft Word, and then manually checked for accuracy. The transcripts were analyzed to identify misconceptions about LLMs in the participants' understanding through a deductive content analysis approach, using the coding scheme in Appendix B.

4.2.3 Tool Usage

After the assessment interview, participants were given access to the mediation tool and were asked to use it during the upcoming week. They were not given any time requirement for how much they should use the tool, but were encouraged to explore it at their own pace and in their own time. The mediation tool was hosted on a virtual machine in UiB's cloud infrastructure service, NREC [43]. Participants were given a link to the tool, as well as a brief introduction to how to access and use it. They were also given contact information of the researcher in case they had any questions or issues with the tool.

4.2.4 Follow-up Interviews

Approximately one week after participants were given access to the mediation tool, they were invited to a follow-up interview. The purpose of this interview was to gather the participants' thoughts and opinions on the mediation tool, as well as to explore if they had any changes in their understanding of LLMs after using the tool. The interview was semi-structured, with a set of prepared questions to guide the conversation, but also allowing participants to freely express their thoughts and opinions. The interviews contained personalized questions based on the participants' responses in the assessment interview, as well as general questions about their experience with the tool. All interviews were conducted in Norwegian. The interview guide can be found in Norwegian in Appendix C. The interviews lasted between 10 and 20 minutes, with an average of around 15 minutes.

The follow-up interviews were transcribed and analyzed in the same way as the assessment interviews, with the transcripts being used for the analysis of the interviews.

4.2.5 Analysis

The interview transcripts were analyzed using deductive content analysis. A predefined coding scheme consisting of the 18 misconceptions identified in chapter 3 was used to code the transcripts. The coding sheet can be found in Appendix B. Each transcript was independently coded by two coders, where the presence or absence of each misconception category was noted. A transcript could be coded with multiple misconceptions. Coders could identify multiple excerpts corresponding to the same misconception within a transcript. However, for the reliability analysis, each misconception category was treated as a binary variable indicating whether the misconception was present or absent in the transcript. Each interview transcript was treated as a single coding unit.

After coding of the assessment interviews, the data was analyzed through IRR analysis to determine the level of agreement between the coders. The analysis was done in a Jupyter Notebook using Python 3.12. The IRR was calculated using percentage of agreement, Cohen's Kappa, calculated with the `cohen_kappa_score` function from the `sklearn.metrics` library [40], and Krippendorff's Alpha, calculated with the `krippendorff.alpha` function from the `krippendorff` library [5]. A IRR coefficient was calculated for each misconception category, treating the presence or absence of the misconception in each transcript as a binary variable. The same analysis was done for the follow-up interviews.

After calculating the IRR-scores, the two coders discussed any discrepancies in their coding and came to a consensus on the final coding for each transcript. The final coding was then used for the analysis of the interviews to examine changes in the presence of misconceptions between the assessment and follow-up interviews.

Chapter 5

Results

5.1 Assessment Interviews

After the two coders completed their assessments, the codings were gathered for IRR analysis. Figure 5.1a shows the calculated agreement percentage, IRR-coefficients and prevalence accross each category. ENVNEGL was not found in any of the transcribt by either coder, so a coefficient could not be calculated. Figure 5.1b shows the level of agreement on each category. It was noticed a weak agreement (both coefficients above 0.6) in the categories of

- COGPASS
- SEMUNDER
- OVERAUTO,

a systematic disagreement (both coefficients below 0) in the categories of

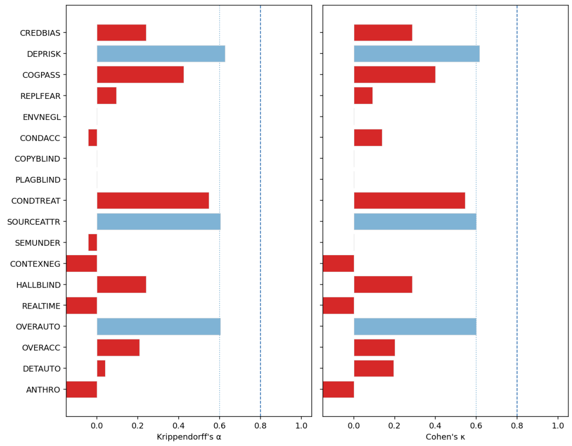
- ANTHRO
- REALTIME
- CONTEXNEG,

a minor systematic disagreement (one coefficient below 0) in the categories of

- SEMUNDER

Category	Agreement	κ	α	Prevalence
ANTHRO	50%	-0.19	-0.267	60%
DETAUTO	50%	0.194	0.040	30%
OVERACC	60%	0.200	0.208	30%
OVERAUTO	80%	0.600	0.604	30%
REALTIME	20%	-0.176	-0.520	80%
HALLBLIND	70%	0.286	0.240	10%
CONTEXNEG	50%	-0.316	-0.267	30%
SEMUNDER	50%	0.000	-0.044	50%
SOURCEATTR	80%	0.600	0.604	30%
CONDTREAT	80%	0.545	0.548	40%
PLAGBLIND	90%	0.000	0.000	10%
COPYBLIND	90%	0.000	0.000	10%
CONDACC	50%	0.138	-0.044	10%
ENVNEGL	100%	—	—	0%
REPLFEAR	60%	0.091	0.095	20%
COGPASS	70%	0.400	0.424	50%
DEPRISK	90%	0.615	0.627	10%
CREDBIAS	70%	0.286	0.240	40%

(a) IRR values for Assessment Interviews



(b) Agreement Coefficients for Assessment Interviews

Figure 5.1: IRR values and kappa/alpha plot for Assessment Interviews

- CONDACC,

and a minor agreement (both coefficients between 0 and 0.6) in the rest of the categories.

The coincidence matrices for each category are shown in Figure 5.2. Here, agreements are shown on the diagonal, while disagreements are shown on the antidiagonal. Notice that all values on the antidiagonal are equal as there are only two coders.

There was no strong or perfect agreement on any of the categories, however the coeffcients are vulnerable to a small dataset. The categories and codings were discussed and it was decided on a final coding matrix to be used for the follow-up interviews. The final coding is shown in Table 5.1.

5.2 Follow Up Interviews

The final coding matrix from the assessment interviews, shown in Table 5.1, was used as a baseline for the coders when coding the follow-up interviews. Misconception deemed present in the assessment interview were flipped to absent if the participants made particular statements showing awareness or understanding of the correct concept. If they did

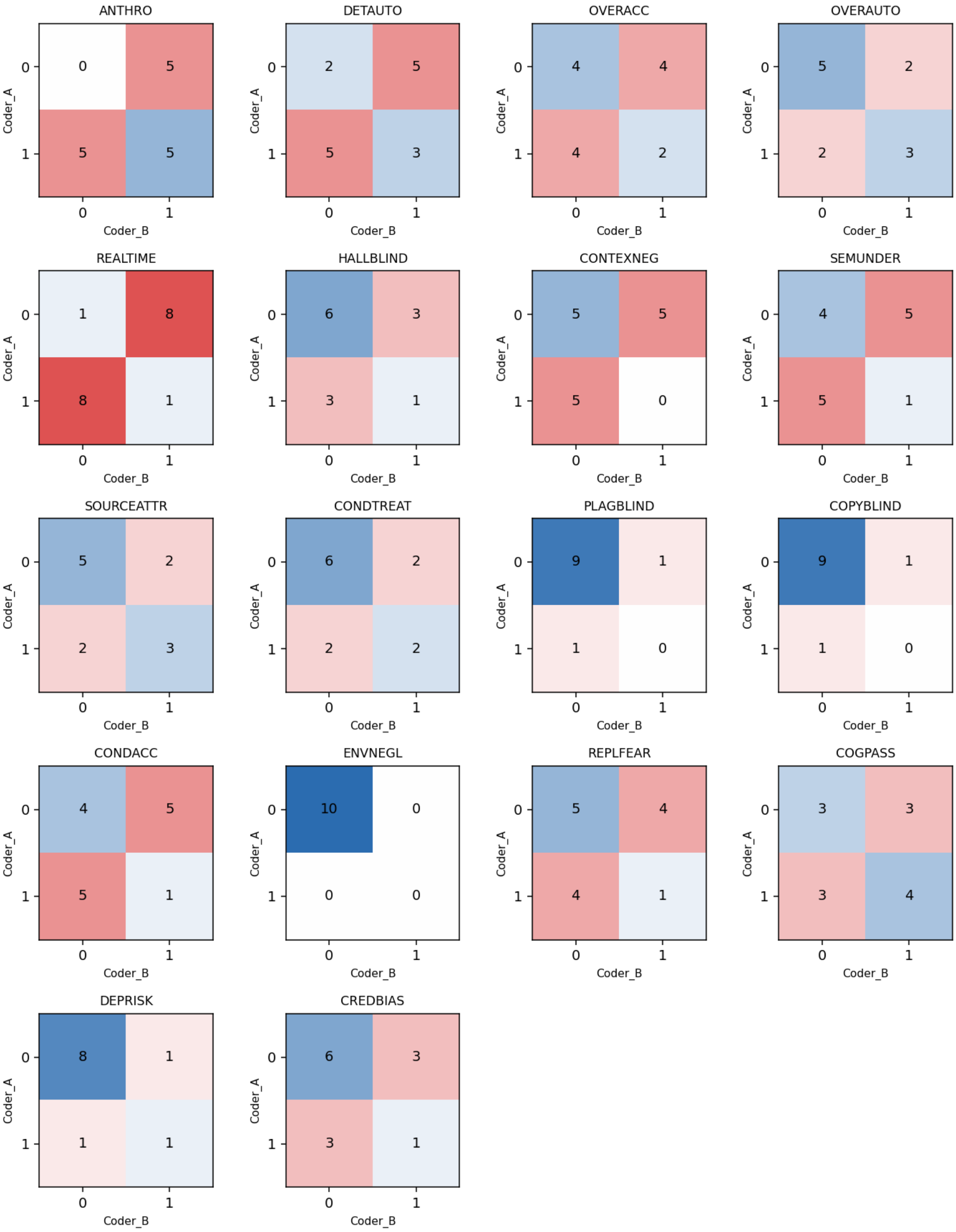


Figure 5.2: Coincidence Matrices for each Category in Assessment Interviews

(a) Dimension 1: FCA

p_id	ANTHRO	DETAUTO	OVERACC	OVERAUTO
1	1	0	1	1
2	1	1	0	0
3	1	1	0	0
4	0	1	0	0
5	1	0	1	0
6	1	0	0	0
7	1	0	1	0
8	1	1	1	1
9	1	0	0	0
10	0	0	0	1

(b) Dimension 2: MU

p_id	REALTIME	HALLBLIND	CONTEXNEG	SEMUNDER	SOURCEATTR	CONDTREAT
1	1	0	0	0	1	0
2	1	0	0	0	0	0
3	1	0	1	0	1	0
4	1	1	1	1	0	0
5	0	1	1	0	1	0
6	1	1	0	0	0	0
7	0	0	0	1	0	0
8	1	0	0	1	1	1
9	1	0	0	1	0	1
10	1	1	0	0	1	0

(c) Dimension 3: EAI

p_id	PLAGBLIND	COPYBLIND	CONDACC	ENVNEGL
1	0	0	0	0
2	0	0	0	0
3	0	0	1	0
4	0	0	0	0
5	0	0	1	0
6	0	0	0	0
7	0	0	1	0
8	0	0	0	0
9	0	0	0	0
10	0	0	1	0

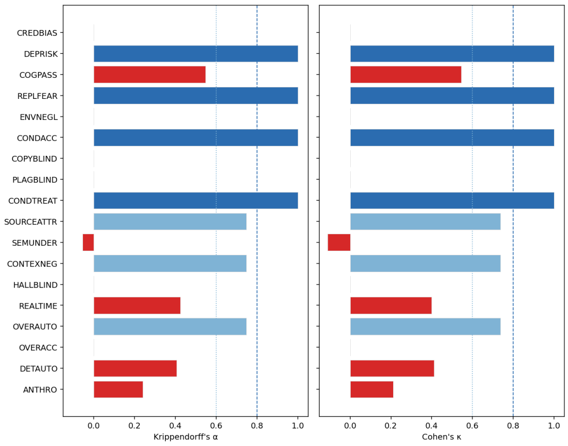
(d) Dimension 4: SCI

p_id	REPLFEAR	COGPASS	DEPRISK	CREDBIAS
1	0	1	1	0
2	0	0	0	0
3	0	0	0	0
4	1	0	0	0
5	0	1	0	0
6	0	1	0	0
7	0	1	0	0
8	0	1	0	0
9	0	0	0	0
10	0	0	0	1

Table 5.1: Binary coding matrix for the assessment interviews, by dimension.

Category	Agreement	κ	α	Prevalence
ANTHRO	70%	0.211	0.240	20%
DETAUTO	80%	0.412	0.406	30%
OVERACC	90%	0.000	0.000	10%
OVERAUTO	90%	0.737	0.747	30%
REALTIME	70%	0.400	0.424	50%
HALLBLIND	90%	0.000	0.000	0%
CONTEXNEG	90%	0.737	0.747	30%
SEMUNDER	80%	-0.111	-0.056	10%
SOURCEATTR	90%	0.737	0.747	20%
CONDTREAT	100%	1	1	20%
PLAGBLIND	100%	—	—	0%
COPYBLIND	100%	—	—	0%
CONDACC	100%	1	1	40%
ENVNEGL	100%	—	—	0%
REPLFEAR	100%	1	1	10%
COGPASS	80%	0.545	0.548	40%
DEPRISK	100%	1	1	10%
CREDBIAS	100%	—	—	0%

(a) IRR values for Follow Up Interviews



(b) Agreement Coefficients for Follow Up Interviews

Figure 5.3: IRR values and kappa/alpha plot for Follow Up Interviews

express the same opinion as they did in the assessment interview, or if they did not provide sufficient evidence to support awareness or viable understanding, the misconception was retained. Figure 5.3 shows the calculated IRR coeffiecients and agreement scores for the coders in the follow up interviews. From the assessment interviews, ENVNEGL, PLAGBLIND, and COPYBLIND were already absent, but in the follow-up coding CREDBIAS were found absent as well. It was found perfect agreement in the categories of

- CONDTREAT
- CONDACC
- REPLFEAR

and sufficient agreement (both coefficients above 0.6) in the categories of

- OVERAUTO
- CONTEXNEG
- SOURCEATTR.

SEMUNDER was the only category with a systematic disagreement (both coefficients below 0). The rest of the categories had a minor agreement (both coefficients between 0 and 0.6). Coincidence matrices for each category are shown in Figure 5.4.

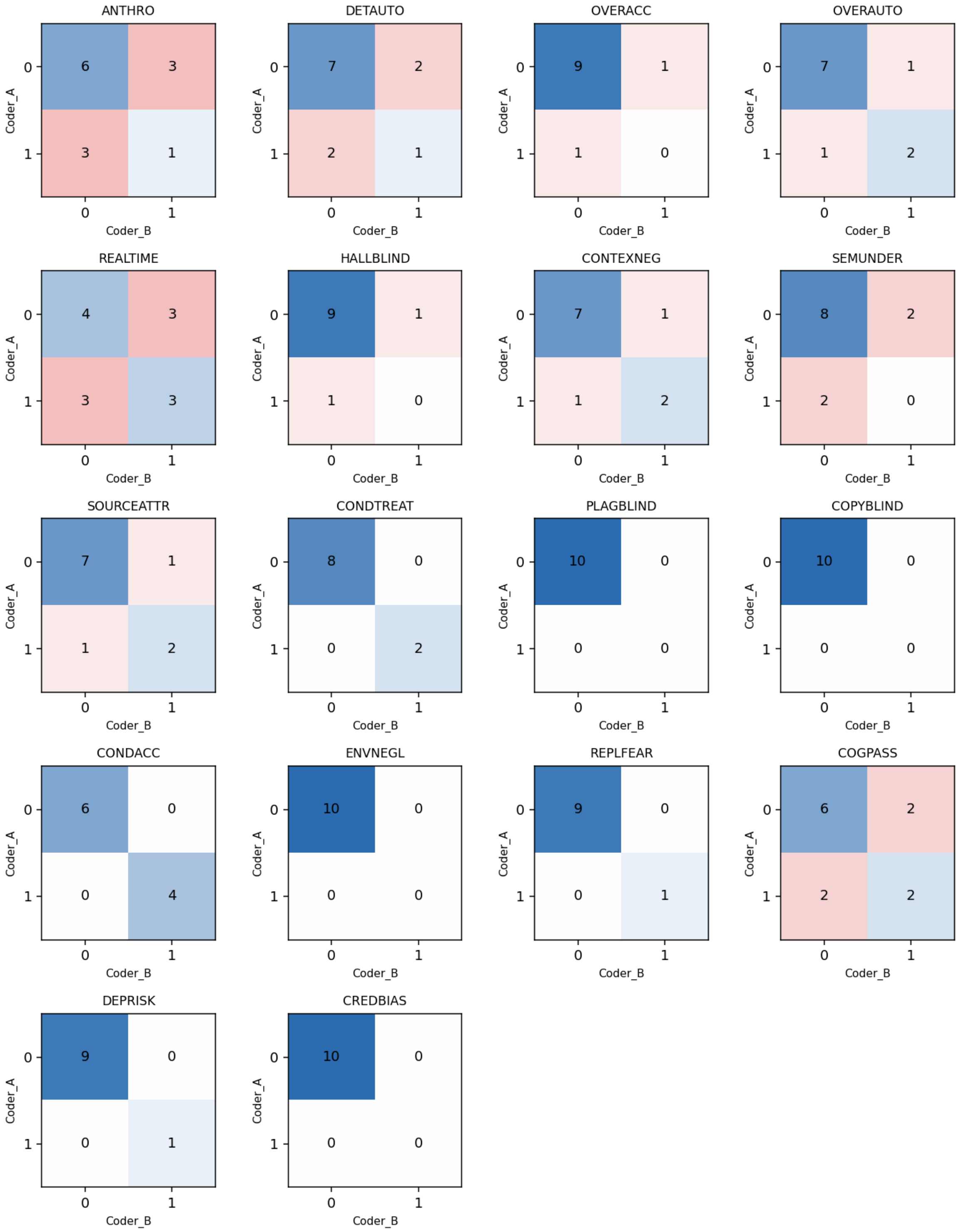


Figure 5.4: Coincidence Matrices for each Category in Follow Up Interviews

The coders met to discuss the coding of the follow-up interviews and agreed on a final coding matrix to be used for the analysis of the results. The final coding matrix is shown in Table 5.2.

5.3 Shifts

(a) Dimension 1: FCA

p_id	ANTHRO	DETAUTO	OVERACC	OVERAUTO
1	1	0	1	1
2	0	1	0	0
3	0	1	0	0
4	0	0	0	0
5	0	0	0	0
6	0	0	0	0
7	0	0	0	0
8	1	0	0	0
9	0	0	0	0
10	0	0	0	1

(b) Dimension 2: MU

p_id	REALTIME	HALLBLIND	CONTEXNEG	SEMUNDER	SOURCEATTR	CONDTREAT
1	1	0	0	0	1	0
2	0	0	0	0	0	0
3	0	0	1	0	1	0
4	1	0	0	0	0	0
5	0	0	1	0	1	0
6	1	0	0	0	0	0
7	0	0	0	0	0	0
8	0	0	0	0	0	1
9	0	0	0	0	0	1
10	1	0	0	0	0	0

(c) Dimension 3: EAI

p_id	PLAGBLIND	COPYBLIND	CONDACC	ENVNEGL
1	0	0	0	0
2	0	0	0	0
3	0	0	1	0
4	0	0	0	0
5	0	0	1	0
6	0	0	0	0
7	0	0	1	0
8	0	0	0	0
9	0	0	0	0
10	0	0	1	0

(d) Dimension 4: SCI

p_id	REPLFEAR	COGPASS	DEPRISK	CREDBIAS
1	0	1	1	0
2	0	0	0	0
3	0	0	0	0
4	1	0	0	0
5	0	1	0	0
6	0	0	0	0
7	0	1	0	0
8	0	1	0	0
9	0	0	0	0
10	0	0	0	0

Table 5.2: Binary coding matrix for the follow-up interviews, by dimension.

Chapter 6

Discussion

6.1 Interpretation of Results

Assessment Interviews

After analyzing the results from the assessment interviews, several key insights emerged. Three out of four categories in the dimension of Ethical and Academic Integrity were declared as absent in the dataset. ENVNEGL was not found in any of the transcripts, by any of the coders, which indicate that the general masses have been exposed to the environmental consequences of GenAI, and are aware of them to some extend. Although most of the participants did not understand why these models are environmentally costly, they were aware of the fact that they are. PLAGBLIND and COPYBLIND were also declared as absent in the dataset, but with initial disagreements between coders. Given the definitions of these categories, the format of a semistructured interview made it difficult to identify these categories. We do not think that these categories are necessarily absent in all participants, but rather that the format of the interview made it difficult to identify them.

Given the small dataset of 10 participants, it was expected that the agreement coefficients would be vulnerable to small changes in the data. In an attempt of isolate the prevalence imbalance from coder disagreements, we compared the calculated κ with the maximum achievable $\kappa_{\max}$. Table 6.1 shows the calculated κ and $\kappa_{\max}$ for each category. Neglecting the non-observed categories, we see that DETAUTO, OVERAUTO, HALLBLIND, SOURCEATTR, CONDTREAT, CONDACC, DEPRISK and CREDBIAS all reach their theoretical maximum ($\kappa = \kappa_{\max}$), indicating that observed agreement is primarily constrained by the distribution rather than coder inconsistencies. Meanwhile, categories exhibiting $\kappa < \kappa_{\max}$ may indicate that the coders had different interpretations of the given definitions.

Table 6.1: Cohen's κ and maximum achievable κ per misconception category (assessment interviews)

Category	κ	$\kappa_{\max}$
ANTHRO	-0.190	0.286
DETAUTO	0.194	0.194
OVERACC	0.200	0.600
OVERAUTO	0.600	0.600
REALTIME	-0.176	0.118
HALLBLIND	0.286	0.286
CONTEXNEG	-0.316	0.737
SEMUNDER	0.000	0.400
SOURCEATTR	0.600	0.600
CONDTREAT	0.545	0.545
PLAGBLIND	0.000	0.000
COPYBLIND	0.000	0.000
CONDACC	0.138	0.138
ENVNEGL	–	–
REPLFEAR	0.091	0.545
COGPASS	0.400	0.800
DEPRISK	0.615	0.615
CREDBIAS	0.286	0.286

The categories with negative IRR coefficients do correspond to the categories were the two coders had different interpretations of the given definitions. It was a consensus between coders that the category of ANTHRO was such a broad concept that it was difficult to apply consistently. The disagreements here were mostly due to missing a specific phrasing or quote in the coding process. The definition of REALTIME was given as "Belief that the LLM retrieves current information." Here, the disagreements came from what counts as "current information." It was settled that this should count as both retrieving and/or information from the internet, the current conversation or other conversations as the interaction happened. For instance, this quote was marked by one of the coders, but was in the end determined to be presence of the misconception:

> "It gets [information] on its server from conversations with other people contiously. And you can also say that one does not know how often the language

> model you are talking with gets new information, or not the language model, but the server you are talking with gets information from other servers."[1]

Modern RAG-based LLMs has also made it more difficult to determine what counts as "retrieving current information", as the line between retrieving and generating has become more blurred.

It was also difference in interpretation of the definition of CONTEXNEG, which was defined as "Failure to recognize limits of LLMs' context window." One interpretation was that the "limits of the context window" should be recognized as the amount of tokens that can be processed at once, while the other interpretation was that it should be recognized as the fact that there is a context window and that limits what and which information the model can use to generate responses, as well as the limitations within that context window. It was settled that the second interpretation should be used, which led to cases like the following being marked as presence of the misconception:

> "Yes, it [the model] usually takes into account what I talked with it about earlier, both in conversations, but also outside of that conversation. It can pull in things from other conversations." [2]

The remaining disagreements were not as systematic as the ones mentioned above, but were rather due to missing a specific phrasing or interpreting a specific phrasing as not sufficient to mark the category as present. These were quickly resolved by going through the disagreements together and discussing the specific cases to land on the consensus shown in Table 5.1.

6.2 Methodological Limitations

Given the amount of categories and the complexity of their definitions, the sample size of 10 participants is too small to make any generalizable conclusions. Optimally also, the analysis of the assessment interviews should also have been completed before starting

[1] Original in Norwegian: "Den får på serveren sin fra samtaler med andre mennesker kontinuerlig. Og så kan du også si at man vet ikke hvor ofte språkmodellen du prater med får inn ny informasjon eller ikke språkmodellen, men den serveren du prater med får informasjon fra andre servere."

[2] Original in Norwegian: "Ja, den bruker å ta hensyn til hva jeg snakket med den om tidligere, både i samtaler, men også utenfor den samtalen. Den kan dra inn ting fra andre samtaler. "

the process of the follow-up interviews, as this would enhance the questions asked during that interview. Due to time limitations it was not possible to do the final coding on the assessment interviews before the follow-up interviews started. With the final coding available, it would be easier to ask tailored questions aimed at the specific categories market for that participant.

6.3 Future Work

Chapter 7

Conclusion

Glossary

Python `https://docs.python.org/3.12/`.

List of Acronyms and Abbreviations

AI Artificial Intelligence.

API Application Programming Interface.

CS Computer Science.

GenAI Generative Artificial Intelligence.

GUI Graphical User Interface.

HAI Human-AI Interaction.

IRR Inter-Rater Reliability.

LLM Large Language Model.

LLMs Large Language Models.

NLP Natural Language Processing.

NLTK Natural Language Toolkit.

RAG Retrieval-Augmented Generation.

REGEX Regular Expression.

UiB The University of Bergen.

Bibliography

[1] Ashok R. Basawapatna, Alexander Repenning, Kyu Han Koh, and Hilarie Nickerson. The zones of proximal flow: guiding students through a space of computational thinking skills and challenges. In *Proceedings of the ninth annual international ACM conference on International computing education research*, pages 67–74, San Diego San California USA, August 2013. ACM. ISBN 978-1-4503-2243-0. doi: 10.1145/2493394.2493404.
URL: https://dl.acm.org/doi/10.1145/2493394.2493404.

[2] Carl T. Bergstrom and Jevin D. West. Modern-day oracles or bullshit machines? how to thrive in a chatgpt world, 2025.
URL: https://thebullshitmachines.com. Online Course.

[3] Steven Bird, Ewan Klein, and Edward Loper. Natural language processing with python, 2009.
URL: https://www.nltk.org/book/. Updated for Python 3 and NLTK 3.

[4] Irina Carnat. Human, all too human: accounting for automation bias in generative large language models. *International Data Privacy Law*, page ipae018, December 2024. ISSN 2044-3994, 2044-4001. doi: 10.1093/idpl/ipae018.
URL: https://academic.oup.com/idpl/advance-article/doi/10.1093/idpl/ipae018/7925747.

[5] Santiago Castro. Fast Krippendorff: Fast computation of Krippendorff's alpha agreement measure. https://github.com/pln-fing-udelar/fast-krippendorff, 2017.

[6] Charlotte Cheatle and Adrian Banks. Accurate but Not Confident or Confident but Not Accurate? Cognitive Offloading Impairs Confidence Calibration in Human-AI Teams. In *Proceedings of the Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems*, pages 1–5, Barcelona , Spain, April 2026.

ACM. ISBN 979-8-4007-2281-3. doi: 10.1145/3772363.3798384.
URL: https://dl.acm.org/doi/10.1145/3772363.3798384.

[7] Sugandha Chhibber, S.R. Rajkumar, and Sandun Dassanayake. Will Artificial Intelligence Reshape the Global Workforce by 2030? A Cross-Sectoral Analysis of Job Displacement and Transformation. *Blockchain, Artificial Intelligence, and Future Research*, 1(1):35–51, February 2025. ISSN 3089-6975. doi: 10.70211/bafr.v1i1.178.
URL: https://journal.wiseedu.co.id/index.php/bafrjournal/article/view/178.

[8] Aeree Cho, Grace C. Kim, Alexander Karpekov, Alec Helbling, Jay Wang, Seongmin Lee, Benjamin Hoover, and Polo Chau. Transformer explainer: Learn how llm transformer models work, 2024.
URL: https://poloclub.github.io/transformer-explainer/. Interactive visualization tool, Georgia Institute of Technology.

[9] Jacob Cohen. A Coefficient of Agreement for Nominal Scales. *Educational and Psychological Measurement*, 20(1):37–46, April 1960. ISSN 0013-1644, 1552-3888. doi: 10.1177/001316446002000104.
URL: https://journals.sagepub.com/doi/10.1177/001316446002000104.

[10] Clara Colombatto and Stephen M Fleming. Folk psychological attributions of consciousness to large language models. *Neuroscience of Consciousness*, 2024(1): niae013, April 2024. ISSN 2057-2107. doi: 10.1093/nc/niae013.
URL: https://academic.oup.com/nc/article/doi/10.1093/nc/niae013/7644104.

[11] Yifan Ding, Matthew Facciani, Amrit Poudel, Ellen Joyce, Salvador Aguinaga, Balaji Veeramani, Sanmitra Bhattacharya, and Tim Weninger. Citations and Trust in LLM Generated Responses, 2025.
URL: https://arxiv.org/abs/2501.01303. Version Number: 1.

[12] Hyo Jin Do, Rachel Ostrand, Justin D. Weisz, Casey Dugan, Prasanna Sattigeri, Dennis Wei, Keerthiram Murugesan, and Werner Geyer. Facilitating Human-LLM Collaboration through Factuality Scores and Source Attributions, 2024.
URL: https://arxiv.org/abs/2405.20434. Version Number: 1.

[13] Yufeng Du, Minyang Tian, Srikanth Ronanki, Subendhu Rongali, Sravan Bodapati, Aram Galstyan, Azton Wells, Roy Schwartz, Eliu A Huerta, and Hao Peng. Context Length Alone Hurts LLM Performance Despite Perfect Retrieval, 2025.
URL: https://arxiv.org/abs/2510.05381. Version Number: 1.

[14] Satu Elo and Helvi Kyngäs. The qualitative content analysis process. *Journal of Advanced Nursing*, 62(1):107–115, April 2008. ISSN 0309-2402, 1365-2648. doi: 10.1111/j.1365-2648.2007.04569.x.
URL: https://onlinelibrary.wiley.com/doi/10.1111/j.1365-2648.2007.04569.x.

[15] Rachel Fertig, Danny Tobey, Gina Durham, and Larissa Bifano. Copyrightability of genai outputs in the us: Key developments, March 2025.
URL: https://www.dlapiper.com/en-us/insights/publications/2025/03/copyrightability-of-genai-outputs-in-the-us-key-developments. Published March 26, 2025.

[16] Luciano Floridi, Jessica Morley, Claudio Novelli, and David Watson. What Kind of Reasoning (if any) is an LLM actually doing? On the Stochastic Nature and Abductive Appearance of Large Language Models, 2025.
URL: https://arxiv.org/abs/2512.10080. Version Number: 1.

[17] Michael Gerlich. AI Tools in Society: Impacts on Cognitive Offloading and the Future of Critical Thinking. *Societies*, 15(1):6, January 2025. ISSN 2075-4698. doi: 10.3390/soc15010006.
URL: https://www.mdpi.com/2075-4698/15/1/6.

[18] Mohammad Ghaseminejad Raeini. The evolution of language models: From N-Grams to LLMs, and beyond. *Natural Language Processing Journal*, 12:100168, September 2025. ISSN 29497191. doi: 10.1016/j.nlp.2025.100168.
URL: https://linkinghub.elsevier.com/retrieve/pii/S2949719125000445.

[19] Kate Goddard, Abdul Roudsari, and Jeremy C Wyatt. Automation bias: a systematic review of frequency, effect mediators, and mitigators. *Journal of the American Medical Informatics Association*, 19(1):121–127, January 2012. ISSN 1067-5027, 1527-974X. doi: 10.1136/amiajnl-2011-000089.
URL: https://academic.oup.com/jamia/article-lookup/doi/10.1136/amiajnl-2011-000089.

[20] Google PAIR. Explorables, n.d.
URL: https://pair.withgoogle.com/explorables/. People + AI Research (PAIR).

[21] Project Gutenberg. Project gutenberg, 2026.
URL: https://www.gutenberg.org/.

[22] Lujain Ibrahim, Canfer Akbulut, Rasmi Elasmar, Charvi Rastogi, Minsuk Kahng, Meredith Ringel Morris, Kevin R. McKee, Verena Rieser, Murray Shanahan, and

Laura Weidinger. Multi-turn Evaluation of Anthropomorphic Behaviours in Large Language Models, February 2026.
URL: `http://arxiv.org/abs/2502.07077`. arXiv:2502.07077 [cs].

[23] Binny Jose and Angel Thomas. Digital anthropomorphism and the psychology of trust in generative AI tutors: an opinion-based thematic synthesis. *Frontiers in Computer Science*, 7:1638657, September 2025. ISSN 2624-9898. doi: 10.3389/fcomp.2025.1638657.
URL: `https://www.frontiersin.org/articles/10.3389/fcomp.2025.1638657/full`.

[24] Artur Klingbeil, Cassandra Grützner, and Philipp Schreck. Trust and reliance on AI — An experimental study on the extent and costs of overreliance on AI. *Computers in Human Behavior*, 160:108352, November 2024. ISSN 07475632. doi: 10.1016/j.chb.2024.108352.
URL: `https://linkinghub.elsevier.com/retrieve/pii/S0747563224002206`.

[25] Klaus Krippendorff. *Content analysis: an introduction to its methodology.* Sage, Thousand Oaks, Calif, 2nd ed edition, 2004. ISBN 978-0-7619-1544-7 978-0-7619-1545-4.
URL: `https://www.metodos.work/wp-content/uploads/2020/05/content_analysis-kippendorf-book.pdf`.

[26] Dennis Kundisch, Jan Muntermann, Anna Maria Oberländer, Daniel Rau, Maximilian Röglinger, Thorsten Schoormann, and Daniel Szopinski. An Update for Taxonomy Designers: Methodological Guidance from Information Systems Research. *Business & Information Systems Engineering*, 64(4):421–439, August 2022. ISSN 2363-7005, 1867-0202. doi: 10.1007/s12599-021-00723-x.
URL: `https://link.springer.com/10.1007/s12599-021-00723-x`.

[27] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks, 2020.
URL: `https://arxiv.org/abs/2005.11401`. Version Number: 4.

[28] Luona Lin and Kim Parker. U.S. Workers Are More Worried Than Hopeful About Future AI Use in the Workplace. February 2025.
URL: `https://www.pewresearch.org/wp-content/uploads/sites/20/2025/02/ST_2025.2.25_AI-Workers_REPORT.pdf`.

[29] Zhicheng Lin. Six Fallacies in Substituting Large Language Models for Human Participants. *Advances in Methods and Practices in Psychological Science*, 8 (3):25152459251357566, July 2025. ISSN 2515-2459, 2515-2467. doi: 10.1177/25152459251357566.
URL: `https://journals.sagepub.com/doi/10.1177/25152459251357566`.

[30] Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the Middle: How Language Models Use Long Contexts. *Transactions of the Association for Computational Linguistics*, 12: 157–173, February 2024. ISSN 2307-387X. doi: 10.1162/tacl_a_00638.
URL: `https://direct.mit.edu/tacl/article/doi/10.1162/tacl_a_00638/119630/Lost-in-the-Middle-How-Language-Models-Use-Long`.

[31] Veronika Lundqvist. Kreativitet i ki-alderen: et opphavsrettslig perspektiv [creativity in the age of ai: a copyright perspective]. *Lov & Data*, December 2025.
URL: `https://lod.lovdata.no/article/2025/12/Kreativitet%20i%20KI-alderen%20et%20opphavsrettslig%20perspektiv`. Published December 11, 2025.

[32] Takuya Maeda and Anabel Quan-Haase. When Human-AI Interactions Become Parasocial: Agency and Anthropomorphism in Affective Design. In *The 2024 ACM Conference on Fairness, Accountability, and Transparency*, pages 1068–1077, Rio de Janeiro Brazil, June 2024. ACM. ISBN 979-8-4007-0450-5. doi: 10.1145/3630106.3658956.
URL: `https://dl.acm.org/doi/10.1145/3630106.3658956`.

[33] Nariman Naderi, Seyed Amir Ahmad Safavi-Naini, Thomas Savage, Mohammad Amin Khalafi, Peter R. Lewis, Zahra Atf, Girish Nadkarni, and Ali Soroush. Across generations, sizes, and types, large language models poorly report self-confidence in gastroenterology clinical reasoning tasks. *npj Gut and Liver*, 3(1): 6, February 2026. ISSN 3004-9806. doi: 10.1038/s44355-026-00053-3.
URL: `https://www.nature.com/articles/s44355-026-00053-3`.

[34] Mahjabin Nahar, Haeseung Seo, Eun-Ju Lee, Aiping Xiong, and Dongwon Lee. Fakes of Varying Shades: How Warning Affects Human Perception and Engagement Regarding LLM Hallucinations, 2024.
URL: `https://arxiv.org/abs/2404.03745`. Version Number: 3.

[35] Paarth Naithani. Issues of Plagiarism and Moral Right to Attribution When Using Work Created by Large Language Models. *Journal of Intellectual Property Rights*,

30(3), 2025. ISSN 09717544, 09751076. doi: 10.56042/jipr.v30i3.11782.
URL: https://or.niscpr.res.in/index.php/JIPR/article/view/11782.

[36] Gabrielle O'Brien, Antonio Pedro Santos Alves, Sebastian Baltes, Grischa Liebel, Mircea Lungu, and Marcos Kalinowski. User Misconceptions of LLM-Based Conversational Programming Assistants, 2025.
URL: https://arxiv.org/abs/2510.25662. Version Number: 2.

[37] Jatin Pandey. Deductive Approach to Content Analysis. In Manish Gupta, Musarrat Shaheen, K. Prathap Reddy, and Madjid Tavana, editors, *Qualitative Techniques for Workplace Data Analysis:*, Advances in Business Information Systems and Analytics, pages 145–169. IGI Global, 2019. ISBN 978-1-5225-5366-3 978-1-5225-5367-0. doi: 10.4018/978-1-5225-5366-3.
URL: http://services.igi-global.com/resolvedoi/resolve.aspx?doi=10.4018/978-1-5225-5366-3.

[38] Norman Paulsen. Context Is What You Need: The Maximum Effective Context Window for Real World Limits of LLMs, 2025.
URL: https://arxiv.org/abs/2509.21361. Version Number: 1.

[39] Marc Pavel, Nenad Petrovic, Lukasz Mazur, Vahid Zolfaghari, Fengjunjie Pan, and Alois Knoll. Hallucination in LLM-Based Code Generation: An Automotive Case Study. In *2025 3rd International Conference on Foundation and Large Language Models (FLLM)*, pages 101–106, Vienna, Austria, November 2025. IEEE. ISBN 979-8-3315-9409-1. doi: 10.1109/FLLM67465.2025.11391125.
URL: https://ieeexplore.ieee.org/document/11391125/.

[40] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

[41] Arthur Perret. A student guide to not writing with chatgpt, November 2024.
URL: https://www.arthurperret.fr/blog/2024-11-14-student-guide-not-writing-with-chatgpt.html.

[42] Madeline G. Reinecke, Fransisca Ting, Julian Savulescu, and Ilina Singh. The Double-Edged Sword of Anthropomorphism in LLMs. In *Online Workshop on Adaptive Education: Harnessing AI for Academic Progress*, page 4. MDPI, February 2025. doi: 10.3390/proceedings2025114004.
URL: https://www.mdpi.com/2504-3900/114/1/4.

[43] Norwegian Research and Education Cloud (NREC), 2026.
URL: https://www.docs.nrec.no.

[44] Sina Rismanchian, Eesha Tur Razia Babar, and Shayan Doroudi. What undergraduate students need to know and actually know about generative AI. *Computers and Education: Artificial Intelligence*, 10:100554, June 2026. ISSN 2666920X. doi: 10.1016/j.caeai.2026.100554.
URL: https://linkinghub.elsevier.com/retrieve/pii/S2666920X26000159.

[45] Grant Sanderson. Large language models explained briefly, November 2024.
URL: https://www.youtube.com/watch?v=LPZh9BOjkQs. YouTube video (3Blue1Brown).

[46] Save the AI. Save the ai, n.d.
URL: https://savethe.ai.

[47] Johannes Schneider. Mental model shifts in human-LLM interactions. *Journal of Intelligent Information Systems*, 63(5):1737–1752, October 2025. ISSN 0925-9902, 1573-7675. doi: 10.1007/s10844-025-00960-6.
URL: https://link.springer.com/10.1007/s10844-025-00960-6.

[48] Thorsten Schoormann, Frederik Möller, and Daniel Szopinski. Exploring Purposes of Using Taxonomies. *Wirtschaftsinformatik2022Proceedings. 5*, 2022.
URL: https://aisel.aisnet.org/wi2022/wi_interdisciplinary/wi_interdisciplinary/5.

[49] Shruthi Shekar, Pat Pataranutaporn, Chethan Sarabu, Guillermo A. Cecchi, and Pattie Maes. People Overtrust AI-Generated Medical Advice despite Low Accuracy. *NEJM AI*, 2(6), May 2025. ISSN 2836-9386. doi: 10.1056/AIoa2300015.
URL: https://ai.nejm.org/doi/10.1056/AIoa2300015.

[50] Bea Stollnitz. How gpt models work, May 2023.
URL: https://bea.stollnitz.com/blog/how-gpt-works/.

[51] Inga Strümke. *Maskiner som tenker*. Kagge, 2023. ISBN 978-82-489-3250-5. OCLC: 1381481881.

[52] Fengfei Sun, Ningke Li, Kailong Wang, and Lorenz Goette. Large Language Models are overconfident and amplify human bias, 2025.
URL: https://arxiv.org/abs/2505.02151. Version Number: 2.

[53] A. M. Turing. Computing Machinery and Intelligence. *Mind*, LIX(236):433–460, October 1950. ISSN 1460-2113, 0026-4423. doi: 10.1093/mind/LIX.236.433.
URL: https://academic.oup.com/mind/article/LIX/236/433/986238.

[54] University of Bergen, Faculty of Social Sciences. Bruk av ki-verktøy for studenter ved sv-fakultetet [use of ai tools for students at the faculty of social sciences]. https://www.uib.no/svf/178548/studenters-bruk-av-kunstig-intelligens-ki-i-eksamensbesvarelser-og-obligatoriske-oppgaver, 2025. Adopted june 10, 2025.

[55] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention Is All You Need, 2017.
URL: https://arxiv.org/abs/1706.03762. Version Number: 7.

[56] Vaishali Vinay. Failure Modes in LLM Systems: A System-Level Taxonomy for Reliable AI Applications, 2025.
URL: https://arxiv.org/abs/2511.19933. Version Number: 2.

[57] Xingyi Wang, Xiaozheng Wang, Sunyup Park, and Yaxing Yao. Mental Models of Generative AI Chatbot Ecosystems. In *Proceedings of the 30th International Conference on Intelligent User Interfaces*, pages 1016–1031, Cagliari Italy, March 2025. ACM. ISBN 979-8-4007-1306-4. doi: 10.1145/3708359.3712125.
URL: https://dl.acm.org/doi/10.1145/3708359.3712125.

[58] Rhiannon Williams. Why Google's AI Overviews gets things wrong. *MIT Technology Review*, May 2024.
URL: https://www.technologyreview.com/2024/05/31/1093019/why-are-googles-ai-overviews-results-so-bad/.

Appendix A

Assessment Interview Guide

Spørsmål til Forhåndsintervju, LLM-explainer

Forklar med egne ord hva AI er og hvordan det fungerer

- Etter dette spørmålet skal AI i denne sammenhengen være forstått som språkmodeller, og ikke AI generelt.
- Dersom det er naturlig, still konkrete oppfølgingsspørmål for å dekke de ulike kategoriene:
  - Hvordan opperer språkmodeller? (OVERAUTO, REALTIME, CONTEXNEG, SEMUNDER, SOURCEATTR, CONDTREAT)
  - Til hvilken grad stoler du på en språkmodells forståelse og svar? (OVERACC, HALLBLIND, SEMUNDER)
  - Hvor kommer informasjonen fra når du bruker en språkmodell, er det noen problemstillinger rundt dette og hvem som eier denne informasjonen? (SOURCEATTR, PLAGBLIND, COPYBLIND)

Forklar hvordan du bruker språkmodeller i hverdagen, og hva du bruker dem til

- Bruk bruksområdene som blir nevnt til å spørre om hvordan denne prosessen fungerer, og hva de tror skjer "bak kulissene" når de bruker språkmodeller.
- Spør om en spesifik prompt de kunne brukt

- Prøv også å finne ut av til hvilken grad de stoler på språkmodeller, og hva de gjør for å verifisere informasjonen de får fra dem. Også om det er noen situasjoner der de ikke stoler på språkmodeller, og hvorfor.

Hvordan påvirker språkmodeller deg i hverdagen, både positivt og negativt?

- Få vedkommende til å drøfte hvilke effekter dette har

Hvordan påvirker språkmodeller samfunnet generelt, både nå og i fremtiden?

- Få vedkommende til å drøfte hvilke effekter dette har, både på kort og lang sikt, og både positivt og negativt.

Appendix B

Coding Sheet

Dimension 1: Foundational Conceptual Accuracy

Anthropomorphism

Attributing human mental states, intentions or understanding to a LLM.

This is a quite broad category, but subjects should be placed in this category if they in some way expressed belief in attributes not fitting of a statistical model and such attributes are aligned with human mental states.

Deterministic Automation

Confusing LLMs with deterministic automation systems.

This could be deterministic chatbot systems, but also more traditional automation systems such as a calculator, or algorithms in general. Subjects who in some way expressed belief that LLMs operate based on deterministic rules, or that they are deterministic systems should be categorized into this category.

Accuracy Overestimation

Overestimating the reliability and correctness of LLM outputs.

Subjects who in some way expressed belief that LLMs are more reliable, correct or trustworthy than they actually are should be categorized into this category.

Autonomy Overestimation

Belief that the LLM operate independently or possess agency.

Subjects who in some way expressed belief that LLMs operate independently, or possess agency should be categorized into this category. This could be belief in some sort of action from the model other than generating output, like "searching the internet".

Dimension 2: Mechanistic Understanding

Real-time Data Belief

Belief that the LLM retrieves current information.

Subjects who in some way expressed belief that LLMs retrieve current information should be categorized into this category. This retrieval can be from the internet, from some database, or from the prompt itself.

Hallucination Blindness

Failure to recognize that LLMs can produce false or fabricated information.

This is not just about recognizing that LLMs can produce false information, but also recognizing that hallucinations is part of the core idea of how LLMs work, and that it is not just a bug or an error that can be fixed.

Context Window Neglect

Failure to recognize limits of LLMs' context window.

If subjects shows no indication of being aware of the limits of LLMs' context window should be categorized into this category. This could be expressed as belief that the model can keep track of an entire conversation, or that it can remember information from previous conversations.

Semantic Understanding

Lack of awareness that the LLM processes language as tokens rather than semantic units.

Subjects who in some way expressed belief that LLMs understand language on a semantic level, or that they read the text as given to them.

Source Attribution Error

Belief that LLM output originate from specific identifiable sources.

Subjects who in some way expressed belief that LLM output originate from specific identifiable sources should be categorized into this category.

Conditional Treatment

Belief that LLMs process prompts differently based on the content or context.

Subjects who in some way expressed belief that LLMs will process one prompt differently than another should be categorized into this category.

Dimension 3: Ethical and Academic Integrity

Plagiarism Blindness

Failure to recognize ethical and legal issues related to the traceability of sources behind LLM-generated content.

This is strongly correlated to the source attribution error, but is more focused on the ethical and legal issues related to the issue. A LLM will mimmic citation patterns in the training data, and failure to recognize the ethical and legal issues related to the traceability of sources behind LLM-generated content should be categorized into this category.

Copyright Blindness

Failure to recognize that LLM-generated content may still be subject to copyright and usage restrictions, despite being machine-generated.

This category is focused on the subjects ability to recognize the questions surrounding ownership of LLM-generated content. That is wether it can be claimed by the prompt creator, the LLM creator, or the original creator of the content in the training data. Failure to recognize this issue should be categorized into this category.

Conditional Acceptance

Ethical acceptance of LLM use only under certain conditions or domains.

Subjects who in some way expressed that "use of LLMs is fine if used for ..." or "LLM use is not fine in my own field, but in ..." should be categorized into this category.

Environmental Neglect

Failure to recognize environmental impact of LLMs.

Subjects who in some way expressed belief that LLMs have negligible environmental impact, or showed lack of awareness of energy consumption associated with LLM training and operation should be categorized into this category.

Dimension 4: Socio-Cognitive Impact

Replacement Fear

Belief that LLMs will replace human roles or professions, expertise, or professions.

Subjects who in some way expressed belief that LLMs will replace human roles or professions should be categorized into this category. Subjects should not be categorized into this category if they only expressed belief that LLMs will replace some tasks within a profession.

Cognitive Passivity

Tendency to rely on LLMs rather than engaging in independent reasoning or critical thinking.

Subjects who, when they described how they use LLMs, showed a tendency to rely on LLMs for tasks that would normally require independent reasoning or critical thinking, should be categorized into this category.

Dependency Risk

Failure to recognize that extensive can lead to long term reliance on LLMs, skill degradation or loss of autonomy.

Subjects who, in describing their use of LLMs, showed no recognition of the risk of becoming reliant on LLMs or described tasks that should not require LLMs but still used them for such tasks, should be categorized into this category.

Credibility Bias

Treating LLM outputs as inherently truthful or authoritative without critical evaluation.

Subjects could think of LLM output as reliable just because of the fact that is machine generated. If this is shown in the subjects description of their use of LLMs, they should be categorized into this category.

Appendix C

Follow-Up Interview Guide

Spørsmål til Oppfølgningsintervju

Hva synes du om verktøyet?

- Hvor lang tid brukte du på det?
- Hvor enkelt var det å bruke det?
- Hvilke funksjoner brukte du mest?
- Hva likte du best med det?
- Hva likte du minst med det?
- Hva kan forbedres med det?
- Hvilke eksterne ressurser brukte du?

Har du fått nye meninger om språkmodeller?

- Har forståelsen din av hvordan språkmodeller fungerer endret seg etter å ha brukt verktøyet?
- Hva tenker du nå om. . . ?
  - Spør spesifikt om misforståelser de hadde før

Har du blitt mer eller mindre skeptisk til teknologien etter å ha brukt verktøyet?

Kommer du til å forandre bruk av språkmodeller etter å ha brukt verktøyet?

- I så fall, hvordan?